MS-PNT-410 Material Performance Standard: Exterior Topcoat System, Waterborne Basecoat / 2K Clearcoat

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This global standard defines what any waterborne basecoat / 2K clearcoat system must pass to be released at any OEM plant, in any color. It names no product, color or plant: those live in the color specification, plant process specification, delivery conditions and approval record. No real company or person is named; all limits are representative.

1. Document control and document set

Field	Value
Standard no.	MS-PNT-410
Revision	C (supersedes Rev B)
Owner	OEM Materials Engineering, Paint and Corrosion (global)
Applies to	All assembly plants worldwide, all colors in this technology
Technology	Waterborne basecoat with 2K solventborne clearcoat, over cathodic e-coat; 3-wet compact process or baked primer-surfacer process
Review cycle	Every 3 years, or on regulatory change
Precedence	Where this standard conflicts with a referenced standard, this standard governs. Where a color specification or plant process specification is stricter, the stricter value governs

This standard is one of five linked documents. Each one varies by a different thing, so no single document is ever written per product.

Document	Example no.	Covers	Written per	Owner
Material performance standard	MS-PNT-410	Performance of the coating technology	Coating technology (global)	Materials Engineering
Color specification	CS-B7M	Master standard, color tolerances, effect, harmony, sensor compatibility	Color (global)	Color & Trim
Plant process specification	PPS-P2-PNT-01	Process parameters, film-build targets, audits, repair loops, local permits	Plant and line	Plant Paint Engineering
Technical delivery conditions	TDC-PNT-01	CoA, batch release, PPAP, change control	Supplier (all its paint materials)	Supplier Quality
Material approval record	MAR-2026-0187	Which supplier product codes are released, against which documents, at which plant	Product x color x plant	Materials Engineering

2. System definition and test conditions

The standard sets allowed ranges only. Each plant sets its own targets inside these ranges in its plant process specification, and each color sets its basecoat target in its color specification.

2.1 Layer build ranges

Layer	Allowed DFT range (µm)	Set in
Cathodic e-coat (reference substrate)	18 to 25	E-coat standard
Functional primer, 3-wet process	15 to 25	Plant process spec

Layer	Allowed DFT range (µm)	Set in
Primer-surfacer, baked process	30 to 40	Plant process spec
Basecoat, solid colors	10 to 20	Color spec
Basecoat, metallic and pearl colors	12 to 18	Color spec
2K clearcoat	40 to 55	Plant process spec

Film thickness per ISO 2808.

2.2 Test panels and reference cure window

Item	Requirement
Substrates	Cold-rolled steel DC04, electro-galvanized DX56D+ZE, hot-dip galvanized DX54D+Z, aluminum AA6016-T4
Pretreatment	Tri-cation zinc phosphate, OEM-released e-coat, 20 min at 175 °C object temperature
Application	Production-equivalent high-speed rotary bells, electrostatic
Low bake	20 min at 130 °C object temperature
Nominal bake	20 min at 140 °C object temperature
Overbake	40 min at 160 °C object temperature
Conditioning	7 days at 23 °C, 50 % RH (ISO 3270)
Panels per test	Minimum 3; all must pass

Every plant's oven window must sit inside this reference window. Adhesion, stone chip and humidity are tested at all three bake conditions; other tests at nominal bake unless stated.

2.3 Reference test colors

A system is approved on four reference colors chosen as worst cases, not on every color. Additional colors are released later with the reduced package in Section 7.

Reference color	Why it is a worst case
Solid black	Scratch and etch visibility, heat build-up
Solid white	Clearcoat yellowing, overbake
Dark aluminum metallic	Intercoat adhesion, flake orientation, stone chip
High-chroma red	Pigment lightfastness in weathering

2.4 Test item index

Every requirement has three parts: the test item (what to check), the test method (how to check it) and the specification (what counts as a pass). This index lists them in one place; the sections hold the full conditions.

Test item	Test method	Specification (summary)	Section
Cross-cut adhesion	ISO 2409	GT 0 to 1	3.2
Pull-off adhesion	ISO 4624	≥ 5 MPa, no intercoat separation	3.2
Gloss	ISO 2813	20°: ≥ 88 GU horizontal, ≥ 85 GU vertical	3.1
Pencil hardness	ISO 15184	≥ F	3.2
Buchholz indentation	ISO 2815	≥ 80	3.2
Scratch resistance	ISO 20566; ASTM D6279	Residual gloss ≥ 60 % (car wash), ≥ 70 % (dry mar)	3.2
Impact resistance	ISO 6272-1	No cracking at 50 cm, 1 kg	3.2
Stone-chip resistance	ISO 20567-1	≤ 1.5 at 23 °C; ≤ 2.0 at -20 °C	3.2
Xenon-arc weathering	SAE J2527	Gloss retention ≥ 80 % at 3000 kJ/m²	4.2
UV fluorescent exposure	ISO 16474-3	Screening only: gloss retention ≥ 80 % at 1000 h	4.2
Salt spray (neutral)	ISO 9227	Scribe creep ≤ 2.0 mm (CRS)	4.1

Test item	Test method	Specification (summary)	Section
Humidity resistance	ISO 6270-2	0(S0); GT ⩽ 1	4.1
Cyclic corrosion	VDA 233-102	Scribe creep ⩽ 3.0 mm (CRS)	4.1
Chemical resistance	ISO 2812-5 and spot tests	Per medium	4.3
Color difference (ΔE) vs master	ASTM E2194; ISO 11664-4; DIN 6175-2	Per color specification	Color spec
Color difference (ΔE) after weathering	ASTM E2194; ISO 11664-4	ΔE*ab ⩽ 1.0 at 45° after 3000 kJ/m²	4.4

3. Appearance and mechanical requirements

3.1 Appearance

Measured on 300 x 500 mm panels sprayed horizontal and vertical.

Property	Method	Horizontal	Vertical
Gloss 20°	ISO 2813	⩾ 88 GU	⩾ 85 GU
Long wave (LW)	Wave-scan structure meter	⩽ 3	⩽ 6
Short wave (SW)	Wave-scan structure meter	⩽ 15	⩽ 20
Distinctness of image (DOI)	Wave-scan structure meter	⩾ 88	⩾ 85
Mottling / clouding (effect colors only)	Visual, 0 to 5 scale (0 = none)	⩽ 1	⩽ 1
Popping limit, clearcoat	Wedge panel, DFT at first pop	⩾ 60 µm	⩾ 60 µm
Sagging limit, clearcoat	Wedge panel with holes	Not applicable	⩾ 55 µm

Property	Method	Horizontal	Vertical
Craters / pinholes	Visual, 300 x 500 mm	None	None
Paint-wetting impairment substances	VDMA 24364, liquid paint	No craters attributable to material	Same

3.2 Mechanical

Test item	Method and conditions	Specification
Cross-cut adhesion	ISO 2409, 2 mm spacing, all bake conditions, all substrates	GT 0 to 1
Pull-off adhesion	ISO 4624, 20 mm dolly, 2K epoxy adhesive, nominal bake and overbake	≥ 5 MPa; no separation between e-coat, primer, basecoat or clearcoat (adhesive or dolly failure accepted)
Pressure water jet adhesion	ISO 16925, X-scribe, 60 °C water, 10 cm, 60 s	Characteristic value ≤ 1
Stone-chip resistance	ISO 20567-1, chilled iron grit 4 to 5 mm, 2 x 500 g at 2 bar; at 23 °C and after conditioning at -20 °C	Rating ≤ 1.5 at 23 °C; ≤ 2.0 at -20 °C; no metal exposed
Stone-chip resistance, gravelometer	SAE J400 at -29 °C (accepted alternative to ISO 20567-1, see Section 6)	≥ 7B; no chips to metal
Pencil hardness	ISO 15184, 750 g	≥ F
Buchholz indentation	ISO 2815, 500 g for 30 s, read within 35 s	Indentation resistance ≥ 80
Impact resistance	ISO 6272-1, 1 kg, 20 mm indenter, direct impact	No cracking or adhesion loss at 50 cm drop height
Scratch resistance, car wash	ISO 20566, 10 cycles	Residual 20° gloss ≥ 60 %; ≥ 80 % after 2 h reflow at 80 °C

Test item	Method and conditions	Specification
Scratch resistance, dry mar	Crockmeter per ASTM D6279, 9 µm abrasive, 10 double rubs	Residual 20° gloss ≥ 70 %

4. Durability and chemical resistance

4.1 Humidity, corrosion and thermal

Run on the full system including e-coat; scribes per ISO 17872, creep d per ISO 4628-8.

Test	Method and conditions	Requirement
Humidity	ISO 6270-2 (CH), 240 h at 40 °C	Blistering 0(S0); GT ≤ 1 at 1 h and 24 h; ΔE*ab ≤ 0.5 at 45°; gloss loss ≤ 5 GU
Water immersion	ISO 2812-2, 240 h at 40 °C	No blisters; GT ≤ 1 after 24 h
Neutral salt spray	ISO 9227, 1000 h	Creep ≤ 2.0 mm (CRS), ≤ 1.0 mm (galvanized)
Cyclic corrosion	VDA 233-102, 6 cycles	Creep ≤ 3.0 mm (CRS), ≤ 1.5 mm (galvanized)
Filiform (aluminum)	ISO 4623-2, 1000 h	Max filament ≤ 2 mm; mean ≤ 1 mm
Thermal cycling	10 cycles: 4 h at -30 °C, 4 h at 80 °C, 16 h at 40 °C / 95 % RH	No cracking or blistering; GT ≤ 1
Heat aging	500 h at 90 °C	No cracking; ΔE*ab ≤ 1.0 at 45°; gloss retention ≥ 90 %

4.2 Weathering

Series release can be granted on accelerated results; outdoor exposure continues after release and the release is withdrawn if a checkpoint fails. Regional exposure sites are in Section 6.

Test item	Method and conditions	Checkpoints	Specification
Xenon-arc weathering	SAE J2527, 0.55 W/m² at 340 nm, 3000 kJ/m²	1000, 2000, 3000 kJ/m²	Gloss retention ≥ 80 %; no cracking, chalking or delamination; color per Section 4.4
UV fluorescent exposure (screening only)	ISO 16474-3, UVA-340 lamps, 8 h UV at 60 °C / 4 h condensation at 50 °C, 1000 h	500, 1000 h	Gloss retention ≥ 80 %; no cracking or chalking. Used to screen formulation changes; never replaces xenon or outdoor results
Outdoor, subtropical	SAE J1976, Florida, 5° south, black box	1, 2, 3, 5 years	2 yr: gloss retention ≥ 85 %. 5 yr: ≥ 70 %; no cracking or delamination; color per Section 4.4
Environmental etch	Florida summer season, 14 weeks, horizontal	End of season	Etch rating ≤ 4 on OEM 0 to 10 scale; not worse than current released clearcoat

4.3 Chemical and fluid resistance

Judged 24 h after exposure on low-bake panels.

Medium	Method	Requirement
Deionized water	ISO 2812-5 gradient oven, 30 min	No damage below 65 °C
Sulfuric acid 1 %	Gradient oven	No damage below 50 °C
Sodium hydroxide 1 %	Gradient oven	No damage below 50 °C
Pancreatin 50 % (bird droppings)	Gradient oven	No damage below 45 °C

Medium	Method	Requirement
Tree resin	Gradient oven	No damage below 50 °C
Gasoline E10	5 drops, evaporate, 3 cycles	No marks; hardness drop ≤ 1 pencil grade
Diesel, washer fluid, coolant	Covered, 24 h	No visible change
Brake fluid DOT 4	1 h, wipe	Full recovery in 24 h
MEK (cure check)	ASTM D5402, 1 kg	≥ 100 double rubs

4.4 Color difference after weathering

Color change after weathering is measured against an unexposed reference panel, not against the color master. Each exposure checkpoint and angle has its own limit.

Field	Value
Test item	Color difference after weathering
Measurement method	Multi-angle spectrophotometer per ASTM E2194; ΔE*ab per ISO 11664-4
Exposures	Xenon arc (SAE J2527) and outdoor Florida (SAE J1976), Section 4.2
Reference	Unexposed panel from the same spray run, stored dark at 23 °C
Parameters	Angles 15°, 25°, 45°, 75°, 110° (-15° recorded for information); D65 / 10°; ΔL*, Δa*, Δb*, ΔE*ab
Unit	ΔE*ab, dimensionless CIELAB units
Qualitative specification	No visible fading, hue shift or chalking vs reference in a D65 light booth

Exposure and checkpoint	ΔE*ab max at 45°	ΔE*ab max at 15°
Xenon 1000 kJ/m²	0.5	0.8

Exposure and checkpoint	ΔE*ab max at 45°	ΔE*ab max at 15°
Xenon 2000 kJ/m²	0.8	1.2
Xenon 3000 kJ/m²	1.0	1.5
Florida 2 years	1.0	1.5
Florida 5 years	2.0	2.5

These are generic limits for all colors. A color specification may set tighter limits per angle, which then govern.

5. Repair and compatibility

These are the generic repair cases every system must survive. Which loops a plant actually runs, and how many times, is set in its plant process specification.

Case	Build and cure	Requirement
Recoat, sanded	Original at overbake; sand P1500; full basecoat + clearcoat; nominal bake	GT ⩽ 1; humidity 240 h pass; stone chip ⩽ 2.0
Recoat, unsanded	Original 3x overbake; basecoat + clearcoat; nominal bake	GT ⩽ 1; pressure water jet ⩽ 1
Low-bake refinish	Supplier-released 2K refinish, 30 min at 60 °C	GT ⩽ 1
Spot polish	Sand P2000 then P3000, machine polish	Gloss ⩾ 95 % of original; no haze
Pipe-loop circulation	4 weeks at plant shear rate	No settling or flocculation; properties within the approval record
Trim tapes and adhesives	OEM-released acrylic foam tape and PU adhesive, 72 h, then humidity 240 h	Cohesive failure in tape or adhesive
Transport protection film	OEM-released film, 2 weeks at 80 °C	No imprint, gloss change ⩽ 3 GU, no residue

6. Material compliance and regional annex

Performance limits are the same worldwide. What changes by region is regulation, the climate the paint must be proven in, and which national test methods are accepted as equivalents.

6.1 Global material requirements

- **Restricted substances:** OEM global restricted substance list applies; no intentionally added lead, cadmium, mercury or hexavalent chromium; SVHC above 0.1 % w/w declared.
- **PFAS:** full disclosure of intentionally added PFAS, with a phase-out plan if present.
- **Material data:** IMDS entry accepted before PPAP; SDS in the language of each producing plant.
- **Carbon footprint:** cradle-to-gate footprint per ISO 14067, updated annually.

6.2 Regional annex

Region	Regulation	Additional climate validation	Accepted method equivalents
Europe	REACH (EC 1907/2006); ELV Directive 2000/53/EC or successor regulation; plant VOC limits under the Industrial Emissions Directive	Florida exposure (Section 4.2) is sufficient	ISO, DIN and VDA methods as written
North America	TSCA; EPA 40 CFR Part 63 Subpart IIII for plant emissions; California Proposition 65 labeling	Additional desert exposure, Arizona, 2 years: gloss retention ≥ 80 %, ΔE*ab ≤ 1.5	SAE J400 accepted for ISO 20567-1; ASTM D3359 for ISO 2409; ASTM B117 for ISO 9227
China	GB 24409-2020 (hazardous substances and VOC in vehicle coatings); GB/T 30512 (prohibited substances in vehicles); local plant emission standards	Additional humid-tropical exposure, Hainan, 2 years; dry-heat exposure, Turpan, 1 year: gloss retention ≥ 80 %, ΔE*ab ≤ 1.5	GB/T 9286 for ISO 2409; GB/T 1771 for ISO 9227; GB/T 1865 accepted with correlation data to SAE J2527

A system approved in one region is released in another only after that region's added validation is complete or running with an interim report.

7. Approval types and test packages

A full system approval happens once per technology and supplier. Most day-to-day work is the smaller packages below it.

Approval type	Trigger	Test package	Documents updated
Full system approval	New supplier or new technology	Sections 3 to 6 on all four reference colors, 3 production-scale batches	Approval record; TDC

Approval type	Trigger	Test package	Documents updated
New color release	New color in an already approved system	Color, appearance, humidity + adhesion, stone chip at overbake, xenon 2000 kJ/m², sensor tests where relevant; outdoor exposure started	Color spec; approval record
Plant introduction	Approved system and color moved to a new plant or line	Line trial, film build, color and appearance audit, repair loops of that plant	Plant process spec; approval record
Requalification	Annually, per supplier	Matrix in TDC	Approval record
Change approval	Any supplier change notified under the TDC	Package set by Materials Engineering based on the change	Approval record